

Software Test Plan (STP)

Project: MedusaJS E-Commerce Platform

Version: 1.0

Date: October 2025

1. Introduction

1.1 Purpose

This Test Plan defines the strategy, objectives, scope, approach, and resources required for testing the MedusaJS E-Commerce Platform. It ensures that all functional and non-functional requirements from the SRS are verified through systematic manual and automated testing.

1.2 Objectives

- Validate that the platform meets functional requirements (registration, checkout, etc.).
- Ensure APIs and UI components perform reliably under expected loads.
- Detect defects early through static and dynamic testing.
- Verify security, usability, and performance compliance.

1.3 References

- Software Requirements Specification (SRS) v1.0
 - MedusaJS Documentation – <https://docs.medusajs.com>
 - GitHub Repository – <https://github.com/medusajs/medusa>
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2. Scope

2.1 In-Scope

Testing will cover:

- Storefront (Customer) UI
- Admin Dashboard (Merchant)
- MedusaJS APIs (/store and /admin)
- Authentication, Product, Cart, Order, and Payment modules

2.2 Out of Scope

- Third-party payment provider internal logic
 - External integrations (e.g., shipping APIs)
 - Browser compatibility for obsolete versions (IE, Safari < v13)
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3. Test Items

Component	Description	Test Type
Authentication	Register, Login, Password Reset	Functional
Product Management	Add, Edit, Delete, Search Products	Functional, Integration
Cart & Checkout	Add to Cart, Payment, Order Creation	Functional, Regression
Orders	Order Tracking, Status Update	Functional
Reports	Export Sales Reports	Functional
APIs	Store, Admin Endpoints	API, Load, Security

4. Test Approach

4.1 Testing Levels

Level	Description
Unit Testing	Conducted by developers on individual modules.
Integration Testing	Verify data flow between backend APIs and frontend UI.
System Testing	Validate end-to-end business workflows.

4.2 Testing Types

Type	Purpose
Functional Testing	Validate all functional requirements.
UI Testing	Verify design consistency and usability.
API Testing	Validate endpoints, parameters, and responses.
Static Testing	Review documentation, code structure, and linting results.

4.3 Test Design Techniques

- **Equivalence Partitioning** (for inputs like email, password)
 - **Boundary Value Analysis** (e.g., quantity limits, price filters)
 - **Decision Table Testing** (for discount and shipping conditions)
 - **Error Guessing** (invalid login, expired token)
 - **Exploratory Testing** (UI and UX testing)
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5. Test Environment

Component	Specification
Frontend	React + TailwindCSS
Backend	Node.js (MedusaJS)
Database	PostgreSQL
Tools	Postman, selenium
Browsers	Chrome, Firefox, Edge
OS	Windows, macOS, Linux
Network	HTTPS enabled, stable internet connection

6. Test Deliverables

Deliverable	Description
Test plan	This document
Test Scenarios & Test cases	Detailed steps for each requirement
Test Data	Sample data for execution
Test Execution Report	Results summary of all test runs
Defect Report	Logged issues with severity and status
Traceability Matrix	Links SRS requirements to test cases

7. Test Schedule

Activity	Start Date	End Date
Requirement Review	Oct 20, 2025	Oct 25, 2025
Test Case Design	Oct 26, 2025	Oct 30, 2025
Environment Setup	Oct 28, 2025	Oct 30, 2025
Test Execution	Nov 1, 2025	Nov 10, 2025
Defect Retesting & Regression	Nov 11, 2025	Nov 15, 2025
Test Closure	Nov 16, 2025	Nov 18, 2025

8. Entry and Exit Criteria

Entry Criteria

- Approved SRS and Test Plan
- Stable build available for testing
- Required test data prepared

Exit Criteria

- All high and medium severity defects fixed
- All functional and regression tests passed

- Test coverage $\geq 90\%$
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9. Risk Management

Risk	Impact	Mitigation
Unstable builds	High	Use staging environment and version control
Missing test data	Medium	Prepare data scripts in advance
API downtime	High	Mock APIs for testing continuity
Late requirement changes	Medium	Implement change control process

11. Tools

- **Test Management:** Jira
- **Automation:** Selenium
- **API Testing:** Postman
- **Version Control:** GitHub